

When Does Structure Help a Magnitude-Only Channel Estimator? Aperture, Effective Rank, and a Prospective Test

Abdullah Haatim

Abstract—A receiver that measures only field magnitude, with a known additive reference, makes multi-user channel estimation a biased phase-retrieval problem whose Fisher information is rank one per measurement and block diagonal across receive elements. Normalised per-element information therefore does not grow with array size, and exploiting aperture requires cross-element structure. On a uniform linear array the Hankel lifting of each channel column supplies exactly such structure, with a capacity of $\lceil N/2 \rceil$ ranks. This predicts, before any measurement, that a rank prior helps only when the channel occupies a small fraction of that capacity — and therefore that it must *hurt* at small aperture. Both follow: enforcing the structure inside an exact expectation-maximization Gerchberg-Saxton iteration, with the order chosen from a held-out pilot residual, gives +2.46 to +3.05 dB at $N = 32$ and -0.19 dB at $N = 8$, while the unstructured baseline moves by 0.012 dB. The governing variable is not path count but effective rank relative to capacity; we use that relation to predict the gain on two channel distributions it was not fitted to, to errors of 0.13 and 0.315 dB. Simulation only; the bounds do not govern biased estimators.

Index Terms—Rydberg atomic receiver, channel estimation, phase retrieval, Hankel matrix, constrained Cramér-Rao bound, effective rank.

I. INTRODUCTION

ATOMIC receivers based on Rydberg states read out radio fields through an optical transition whose response is proportional to field *magnitude*, discarding phase [1], [2]. Multi-user channel estimation is therefore biased phase retrieval, and Gerchberg-Saxton-type alternating projection [3] is the natural solver; recent work unrolls it into a trained network [4], and a rank projection has been placed inside projected gradient descent for the same problem [5].

The atomic receiver enters this paper only through the observation model — a magnitude-only measurement with a known additive reference, (1). Nothing below depends on the vapour-cell physics, so the results apply to any array receiver with that measurement, of which the Rydberg receiver is the motivating instance.

This paper is organised around a single argument, stated first and tested afterwards.

Step 1: unstructured solvers cannot use aperture. Under the unconstrained row-wise parameterisation of (1), row n of **GS** enters only measurement row n , so the Fisher information is block diagonal across receive elements (§III). The per-element estimation problem is unchanged by adding elements,

and normalised per-element information does not grow with N .

Step 2: aperture is therefore usable only through cross-element structure. On a ULA each path contributes one spatial complex exponential, so the Hankel lifting of every channel column is rank deficient [6], [7]. That is one such structure, not the only one.

Step 3: the structure has a finite capacity. The lifting admits at most $r_{\max}(N) = \lceil N/2 \rceil$ ranks (§II). A rank prior is informative only when the channel occupies a small fraction of it.

Step 4: so the prior must fail, and then harm, at small aperture. At $N = 8$ the capacity is 4 while the simulated path counts are $L_k \sim \mathcal{U}\{3, \dots, 7\}$; the constraint is frequently vacuous and, when active, truncates genuine components. A *sign reversal at small N is predicted by the argument, not discovered afterwards*, and §IV-B reports it as such.

The remaining difficulty is knowing *when* “a small fraction of capacity” holds. A qualitative appeal to sparsity is unusable: a 16-path channel on a 32-element array is algebraically full rank yet has effective rank 8.73, and its lifting is compressible with nothing sparse about it. §IV-D replaces path count with effective rank relative to capacity, and §IV-E tests that replacement prospectively.

Contributions.

- 1) The information argument above, with a Rician Cramér-Rao bound in which each magnitude measurement contributes *rank-one* information, and a geometry-constrained bound in tangent-space form because the parameterisation is rank deficient.
- 2) The aperture-scaling result, presented as a test of a prediction the argument makes in advance.
- 3) Replacement of the path-count heuristic by $r_{\text{eff}}/r_{\max}$, with the capacity $r_{\max} = \lceil N/2 \rceil$ derived, and a measurement separating what the *prior* does from what the *estimator* does.
- 4) Two cross-generator predictions, each registered before measurement.

II. SYSTEM MODEL AND ESTIMATOR

An N -element ULA serves K single-antenna users over P pilots. With $\mathbf{G} \in \mathbb{C}^{N \times K}$, pilots $\mathbf{S}$, known reference beam $\mathbf{B}$ and noise $\mathbf{W}$, the receiver observes

$$\mathbf{Z} = |\mathbf{GS} + \mathbf{B} + \mathbf{W}|, \quad \mathbf{W}_{np} \sim \mathcal{CN}(0, \sigma^2), \quad (1)$$

A. Haatim is with the Department of Electrical and Electronics Engineering, Birla Institute of Technology and Science, Pilani — Pilani Campus, India (e-mail: f20220519@pilani.bits-pilani.ac.in).

the modulus applied after the noise, so (1) cannot be linearised. Each user's channel is a sum of L_k plane waves on the array manifold,

$$g_k[n] = \sum_{\ell=1}^{L_k} \alpha_{\ell,k} e^{-j(n-1)\psi_{\ell,k}}, \quad \psi_{\ell,k} = \pi \sin \theta_{\ell,k}, \quad (2)$$

with $L_k \sim \mathcal{U}\{3, 7\}$ [5] and $\theta \sim \mathcal{U}(-90^\circ, 90^\circ)$. The reference-to-signal ratio (RSR) is the reference power over the *single-user* received power.

Structure and capacity. Substituting (2) into the $(N-p) \times (p+1)$ Hankel lifting $\mathcal{H}_p$ separates the exponential in the two indices, so each path contributes one rank-one outer product and $\text{rank } \mathcal{H}_p(\mathbf{g}_k) \leq L_k$. Since $\mathcal{H}_p$ is $(N-p) \times (p+1)$,

$$r_{\max}(N) = \max_p \min(N-p, p+1) = \lceil N/2 \rceil, \quad (3)$$

and if $L_k \geq r_{\max}(N)$ the constraint is vacuous. K **cancels**: the structural model uses $3 \sum_k L_k$ parameters against $2NK$ unstructured, and the ratio is independent of K , so the gain should be K -invariant at fixed pilot adequacy — tested in §IV-F.

The estimator. HS-GS applies, after every exact EM-GS update, a Cadzow projection $\mathcal{P}_r(\mathbf{g}) = \mathcal{H}_p^\dagger(\Pi_r(\mathcal{H}_p(\mathbf{g})))$, where Π_r truncates to r leading singular values and $\mathcal{H}_p^\dagger$ averages antidiagonals. One application is *not* idempotent — it is one step of an alternating projection between the rank set and the Hankel set — so the sweep count is part of the operator's definition. All results below use $n_{\text{cz}} = 4$ sweeps, $T = 100$ iterations and a selection budget of 25. With the projection disabled HS-GS reproduces EM-GS to $\max |\text{diff}| = 0$, so the projection is provably the only difference.

Order selection uses no oracle. A fraction $\nu = 0.3$ of the pilot columns is held out; each candidate $r \in \{1, \dots, r_{\max}\}$ is scored by the held-out magnitude residual and the minimiser is used. The in-sample residual cannot serve: constraining $\mathbf{G}$ can only worsen the fit to the pilots used for fitting, so an in-sample criterion always prefers the largest candidate. When the selector returns $\hat{r} \geq r_{\max}$ the projection is the identity and the estimator *abstains*, falling back exactly to EM-GS. This is a designed behaviour and §IV-D shows it is load-bearing.

Why not estimate the angles directly? Subspace methods such as root-MUSIC and ESPRIT recover $\{\psi_{\ell,k}\}$ from the signal subspace, but that subspace requires either multiple snapshots or the complex field. With magnitude-only single-snapshot data it is identifiable only *after* phase retrieval has succeeded, at which point the channel is already estimated. We therefore regularise toward the exponential-sum manifold rather than parameterising it.

Related work. Structured low-rank Hankel recovery for sums of exponentials — EMaC, ALOHA, LORAKS — recovers such a matrix from *partial linear* observations, with convex relaxations and recovery guarantees. **[TODO: VERIFY CITATION]** Gridless line-spectrum and atomic-norm methods likewise assume linear measurements and a separation condition. **[TODO: VERIFY CITATION]** Toeplitz and Vandermonde covariance methods in array processing impose structure on a covariance accumulated over many snapshots. **[TODO: VERIFY CITATION]** All three differ

from the present setting in the same way: our measurement is a *modulus*, so the map is not linear, no convex relaxation applies, and the projection is used as a regulariser inside a non-convex iteration rather than as the estimator. We claim no new projection algorithm; Cadzow's is used as given.

Statistics. The primary statistic is the paired per-trial median within an SNR bin, with bootstrap 95% intervals. Where a curve is compared to a bound we use the ratio of summed errors, which is what a bound on the mean-squared error governs; both are reported for the headline cell.

III. PERFORMANCE BENCHMARK

A. Rank-one Fisher information

Conditional on $\mathbf{G}, \mathbf{S}, \mathbf{B}$ the noiseless field at (n, p) is a constant λ , so $z = |\lambda + w|$ is Rice distributed and its density depends on λ *only through* $a = |\lambda|$. With $\frac{d}{dx} \log I_0(x) = R(x)$ and $\kappa = 2za/\sigma^2$, the score is $\partial_a \log p = \frac{2}{\sigma^2} [zR(\kappa) - a]$; writing $\beta = \sigma^{-4}(\mathbb{E}[z^2 R^2(\kappa)] - a^2)$, one measurement carries scalar information 4β , so in real coordinates $\boldsymbol{\eta} = [\Re \text{vec } \mathbf{G}; \Im \text{vec } \mathbf{G}]$,

$$\mathbf{J}_n = \sum_{p=1}^P 4\beta(a_{n,p}, \sigma^2) \mathbf{u}_{n,p} \mathbf{u}_{n,p}^T. \quad (4)$$

Each magnitude measurement contributes rank-one information, not rank two. Crediting both quadratures overstates it and gives a bound too low by 0.17–4.64 dB across our 36 points. An identifiability check settles the form: at $K = 3$, $P = 3$ the per-row problem is unidentifiable and (4) is correctly singular, whereas the rank-two construction claims full rank.

Because row n of $\mathbf{GS}$ enters only measurement row n , $\mathbf{J}$ is *block diagonal across receive elements*. **This is Step 1 of the argument**, and it is why an unstructured estimator — and any estimator attaining the unconstrained bound — is flat in N under a normalised metric.

B. Constrained bound on the geometric manifold

The unconstrained bound governs estimators treating $\mathbf{G}$ as $2NK$ free parameters. HS-GS exploits (2), whose parameters $\boldsymbol{\phi} \in \mathbb{R}^{3 \sum_k L_k}$ number about 45, *independent of* N . The textbook form $\mathbf{D}(\mathbf{D}^T \mathbf{J} \mathbf{D})^{-1} \mathbf{D}^T$ with $\mathbf{D} = \partial \boldsymbol{\eta} / \partial \boldsymbol{\phi}$ is unusable, $\mathbf{D}$ being rank deficient: at $N = 8$ its mean numerical rank is 40.4 against ≈ 44.7 parameters, and 28.1% of realisations have $3 \sum_k L_k > 2NK$. We use the Gorman–Hero / Stoica–Ng tangent-space form [9], [10] with $\mathbf{U}$ an orthonormal basis of $\text{range}(\mathbf{D})$:

$$\text{CCRB} = \mathbf{U}(\mathbf{U}^T \mathbf{J} \mathbf{U})^{-1} \mathbf{U}^T, \quad (5)$$

which depends on the tangent *space* and is invariant to over-parameterisation. Both bounds average over the same realisations as the estimator curves.

Bias caveat. Both bounds constrain the covariance of *unbiased* estimators. All estimators here run to a fixed budget with a shrinking update, and HS-GS additionally truncates rank, so none is unbiased and neither bound applies strictly. The CCRB is the primary reference in Fig. 1 because it is the one matched to the estimator's model class; the unconstrained bound is plotted as a secondary line and governs EM-GS only.

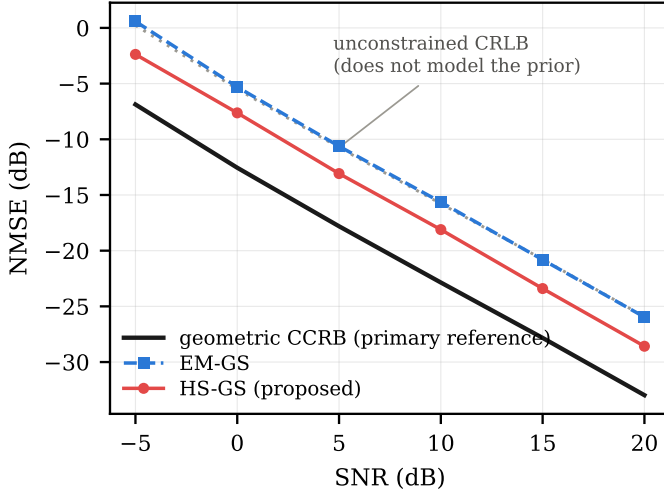


Fig. 1. NMSE at $N = 32$, $K = 3$, $P = 30$, RSR 12 dB, 400 paired trials per point. The geometric CCRB is the primary reference; the unconstrained CRLB is shown faint because it does not model the prior and governs EM-GS only. Neither bound governs a biased estimator.

IV. NUMERICAL RESULTS

Unless stated, $N = 32$, $K = 3$, $P = 30$, RSR 12 dB, 400 paired trials per SNR point, with both estimators handed identical realisations.

A. Main result against the bounds

HS-GS improves on EM-GS by +2.46 to +3.05 dB across the sweep (six-point mean +2.68 dB; ratio-of-sums +2.33 to +2.96 dB), and the projection is active in 100% of trials at this cell (Fig. 1).

The benchmark makes that number interpretable. EM-GS tracks the unconstrained bound to within 0.143 dB for $\text{SNR} \geq 5$ dB, so the bound is informative rather than vacuous. The geometric CCRB lies 7.05–7.11 dB below it at this configuration: that gap is what the structural model is worth in principle, and HS-GS recovers part of it, remaining above the CCRB at all six points by 4.39–4.93 dB.

B. Aperture scaling: the predicted sign reversal

Step 4 predicts the gain turns negative at small aperture. Sweeping $N \in \{8, 16, 32\}$ over 12 operating points each (2 pilot counts $\times$ 6 SNR values), the mean gain is -0.19 , $+0.78$ and $+2.85$ dB. **Under the same averaging EM-GS varies by 0.012 dB** — exactly as §III requires of an estimator whose information is block diagonal in N .

At $N = 8$ the projection is inactive in 42.5% of trials, and where active it truncates genuine components: the gain is $+1.47$ dB at -5 dB SNR but -2.23 dB at high SNR, because truncation bias does not shrink with noise while the variance term does. *This is the predicted failure, measured.* The sign of any scalar summary at $N = 8$ depends on the pooling rule, so we report per-point values.

C. Path count is the wrong coordinate

Holding $N = 32$ and varying only the path count, with L identical across users at $\text{SNR} = 5$ dB, the gain decays monotonically from $+7.04$ dB at $L = 2$ to -0.12 dB at $L = 16 = r_{\max}$, while EM-GS moves by 0.191 dB in total. So the operative variable is channel complexity relative to capacity, and aperture matters only because it sets the capacity.

D. Effective rank, and what the prior does versus the estimator

Path count is still imperfect: clustered or unequal-strength paths occupy fewer effective dimensions. We index by the Roy–Vetterli effective rank [8] of the lifting, $r_{\text{eff}} = \exp(-\sum_i p_i \ln p_i)$ with $p_i = \sigma_i / \sum_j \sigma_j$, normalised by capacity, $\rho = r_{\text{eff}} / r_{\max}$.

Re-indexing the sweep of §IV-C moves the zero crossing from $L/r_{\max} = 0.90$ to $\rho = 0.518$, and the useful region becomes $\rho \lesssim 0.5$ — a statement about any channel rather than one simulator’s L values. Across $N \in \{16, 32, 64\}$ the crossing is 0.588, 0.518, 0.544, a spread of 0.070 over a fourfold change in aperture (Fig. 2).

The crossing belongs to the prior, not to the estimator, and the two must be stated separately. Pushing past the crossing with the proposed estimator never yields a negative: at $\rho = 0.608$ and 0.735 ($N = 16$) it gives $+0.078$ and $+0.000$ dB, and at $\rho = 0.596$ and 0.669 ($N = 64$), $+0.044$ and $+0.018$ dB. The reason is abstention: at $\rho = 0.735$ the selector returns $\hat{r} \geq r_{\max}$ in 38.8% of trials, and where it does the estimator is exactly EM-GS. Forcing the projection active at that same cell gives -0.075 dB, CI $[-0.186, -0.026]$. So the prior does become harmful past the crossing, and the abstention rule is what prevents the estimator from following it down. $\Delta_{\text{HS}} \geq 0$ is a **property of the selector, not evidence that the prior is never harmful.**

The gain magnitude does not collapse onto the same axis. Over the window both arrays cover, $N = 16$ lies above $N = 64$ by a mean of 0.493 dB and a maximum of 0.901 dB, one-signed throughout. *We have no account of it.* The one candidate we tested — that the selector’s rank grid is coarser at small capacity — is refuted: varying the pencil at fixed $N = 32$ and fixed ρ gives a span of 0.390 dB over admissible-rank counts 8 to 16 with *no monotone trend*, and the dependence runs mildly backwards. A pair with opposite matrix shapes but nearly equal rank counts (24×9 and 8×25) gives $+3.199$ and $+3.121$ dB, so the effect tracks the rank count and not the shape.

ρ is a design-time coordinate, not a receiver-observable one, since r_{eff} is computed on the noiseless channel. The selector’s abstention rate is observable and is monotone in ρ at fixed aperture, but it is confounded by capacity: at $N = 64$ a cell at $\rho = 0.501$ abstains in 0.0% of trials while at $N = 16$ a cell at $\rho = 0.371$ abstains in 4.0%, so no single threshold separates the useful region. An observable proxy therefore exists within an aperture but has not been established across apertures.

E. Cross-generator prediction

A relation fitted and evaluated on the same data proves little. We froze the ρ -to-gain relation obtained above, committed

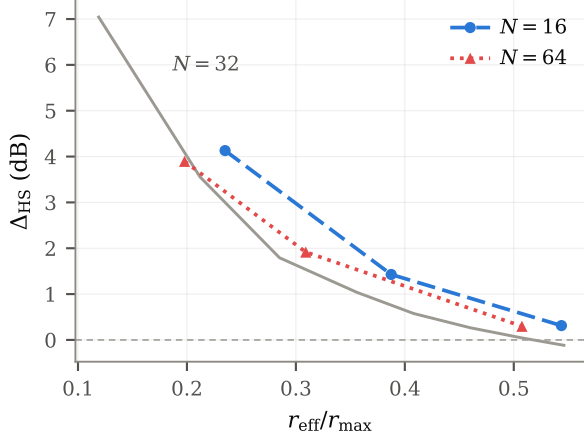


Fig. 2. Δ_{HS} against $\rho = r_{\text{eff}}/r_{\text{max}}$ for $N \in \{16, 32, 64\}$. The boundary location is stable across a fourfold change in aperture; the gain magnitude is not, and that residual is unexplained.

numerical predictions to version control, and only then ran two channel generators that played no part in constructing it. The rule is piecewise-linear interpolation on a fixed eight-point table — no regression, no spline — so it re-executes exactly.

Clustered Saleh–Valenzuela. At $\rho = 0.331$ the relation predicted +1.30 dB; the measurement over 276 trials gave +1.173 dB, CI [+1.017, +1.380] — an error of 0.13 dB.

A second, independently specified configuration. At $\rho = 0.400$ the prediction was +0.648 dB with interval [+0.367, +1.015]; the measurement over 266 trials gave +0.963 dB, CI [+0.808, +1.086] — an error of 0.315 dB, inside the stated interval but on its upper edge.

Scope. These are *cross-generator* tests, not out-of-model tests: both generators still produce channels of the form (2), and only the distribution over $(L_k, \alpha_{\ell,k}, \theta_{\ell,k})$ changes. They show ρ transfers across path distributions, not that it survives a violation of the ULA model.

F. Users, cost, and where the projection belongs

K-invariance. At fixed pilot adequacy $P/2K = 3.33$ the gain varies by 0.094 dB across $K \in \{2, 3, 4\}$ for SNR ≥ 5 dB. Pooled over all SNR the spread is 0.294 dB against a pre-registered falsifier of 0.3 dB — missed by 0.006 dB, and reported rather than dropped.

Cost. Order selection is 57.7–85.5% of runtime. A coarse-to-fine search over a stride-3 grid with local refinement returns the same $\hat{r}$ in 93.3% of trials, cuts candidate evaluations from 16.0 to 9.6 and the selection stage by 1.68 $\times$, at an aggregate cost of 0.083 dB.

Placement is SNR-dependent, and the effect is small. Comparing interleaved projection against a single post-hoc Cadzow projection at the same selected order, interleaving is *worse* at 0 dB (−0.164 dB, CI [−0.317, −0.066]), indistinguishable from +5 to +15 dB, and better only at +20 dB (+0.168 dB, CI [+0.102, +0.246]); the six-point mean is +0.003 dB. Interleaving is therefore not required in this classical iteration, and we do not claim it is — a single

projection recovers nearly all of the benefit at a fraction of the cost.

V. LIMITATIONS AND CONCLUSION

A magnitude-only receiver’s information is rank one per measurement and block diagonal across elements, so aperture is usable only through cross-element structure. The ULA Hankel prior supplies it, with capacity $\lceil N/2 \rceil$; the prior therefore helps only below a fraction of that capacity, and the predicted sign reversal at small aperture is observed. The governing coordinate is $r_{\text{eff}}/r_{\text{max}}$, whose boundary location is stable across a fourfold change in aperture and which predicted the gain on two unseen channel distributions to 0.13 and 0.315 dB.

Limitations. All results are simulation only, with no measured hardware and no recorded channel data. They cover geometric ULA channels of the form (2) at the array sizes, pilot counts and generators tested, and the cross-generator tests change the distribution but not the model family; no claim is made beyond them. The gain magnitude does not collapse onto the proposed axis and its residual of 0.493 dB is unexplained. ρ is not receiver-observable. Two of the three boundary crossings cannot be bracketed from the negative side by the proposed estimator, because abstention prevents it; $\rho \approx 0.75$ is in any case unreachable at $N = 64$, where the path generator fails above $L = 48$ ($\rho \approx 0.67$). We establish no fundamental bound: the CRLB and CCRB govern unbiased estimators and every estimator here is biased.

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